Exploratory Data Analysis

SIMULATED DATA NOTICE: all results in this document are calculated from a synthetic dataset created to test the research pipeline. They are not findings about real students, and the dataset contains no genuine questionnaire responses.

1. Dataset overview and sample composition

1000 simulated students, 17 columns, 60 missing cells (59 rows affected). Category rule: High ≥ 60, Medium 50-59.9, Low < 50 (mean score %, current semester).

Year of study	Count	%
2	261	26.1
3	260	26.0
4	253	25.3
1	216	21.6
5	10	1.0

Performance category	Count	%
Medium	381	38.1
High	355	35.5
Low	264	26.4

Internet access	Count	%
Yes	802	80.2
No	198	19.8

Study location	Count	%
Home	329	32.9
Library	272	27.2
Hostel	200	20.0
Other	106	10.6
Classroom	93	9.3

2. Descriptive statistics

Variable	Mean	SD	Min	Median	Max	Missing	IQR outliers
study_hours_weekly	12.16	7.18	0.0	12.0	33.0	6	1
attendance_percentage	75.66	13.15	30.0	76.0	100.0	11	4
assignment_completion_percentage	77.58	14.74	15.0	78.0	100.0	0	5
sleep_hours	6.6	1.13	3.5	6.6	10.0	14	11
number_of_courses	6.27	1.35	3.0	6.0	10.0	0	0
social_media_hours_daily	3.59	1.61	0.0	3.6	8.2	13	2
previous_performance	56.42	10.27	25.6	56.2	85.2	16	3
current_performance	56.37	10.09	23.3	56.7	88.5	0	9

3. Predictors by performance category

Predictor	Median Low	Median Medium	Median High	Mean Low	Mean Medium	Mean High	Kruskal-Wallis H	p-value
Attendance (%)	71.0	75.0	81.0	70.61	75.2	79.88	69.86	<0.001
Study hours per week	9.0	12.0	15.0	8.98	11.58	15.15	109.42	<0.001
Sleep (hours/night)	6.2	6.5	6.8	6.29	6.61	6.81	33.43	<0.001

Predictor	Median Low	Median Medium	Median High	Mean Low	Mean Medium	Mean High	Kruskal-Wallis H	p-value
Assignment completion (%)	69.0	79.0	83.0	69.75	77.98	82.99	119.69	<0.001
Social media (hours/day)	4.05	3.7	3.3	3.95	3.63	3.28	25.37	<0.001

Kruskal-Wallis tests are descriptive here: the dataset is simulated, so a small p-value only confirms that the pipeline detects the structure built into the simulator.

Cross-tabulations (counts)

Year of study	Low	Medium	High
1	60	83	73
2	75	98	88
3	63	104	93
4	63	93	97
5	3	3	4

Study location	Low	Medium	High
Classroom	26	35	32
Home	81	135	113
Hostel	55	72	73
Library	75	97	100
Other	27	42	37

Internet access	Low	Medium	High
No	60	70	68
Yes	204	311	287

4. Preliminary observations (all calculated from the simulated file)

• Class balance: Low 264 (26.4%), Medium 381 (38.1%), High 355 (35.5%). The largest class is 38.1% of rows, so the simulated data is only mildly imbalanced.
• Study hours: median 9.0 (Low), 12.0 (Medium), 15.0 (High) hours per week; Kruskal-Wallis H = 109.42, p <0.001. The ranges overlap heavily (see box plots), so no category is separable on study hours alone.
• Attendance, assignment completion and sleep show the same ordering (Low < Medium < High in the medians); social media use runs the other way (median 4.05 for Low vs 3.3 for High hours per day).
• Correlations with current performance: assignment completion 0.367, study hours 0.355, attendance 0.303, sleep 0.201, social media -0.164, number of courses 0.021, previous performance 0.643.
• Year of study, study location and internet access showed no clear association with category in this file (chi-square p = 0.932, 0.959 and 0.367 respectively).
• Previous performance is the strongest single correlate of current performance (r = 0.643, n = 984); see the redundancy check in the cleaning report.
• On a single stratified 80:20 split (test n = 200), test macro-F1 was 0.184 for the majority baseline, 0.556 for Logistic Regression and 0.549 for Random Forest. Cross-validated macro-F1 on the training part was 0.570 (SD 0.021) and 0.580 (SD 0.017). The gap between the two models is smaller than the fold-to-fold spread, so neither is shown to be better. These runs only prove the pipeline works.

5. Limitations

• The data is synthetic; relationships were put there by the simulation design, so none of these patterns is evidence about real students.
• Real data will be smaller (planned 50-150 students), self-reported and voluntary, so expect more noise, selection bias and measurement error.
• Different university grading scales cannot be assumed equivalent; real data needs a documented mapping.

- Association and prediction do not establish causation.

6. Charts

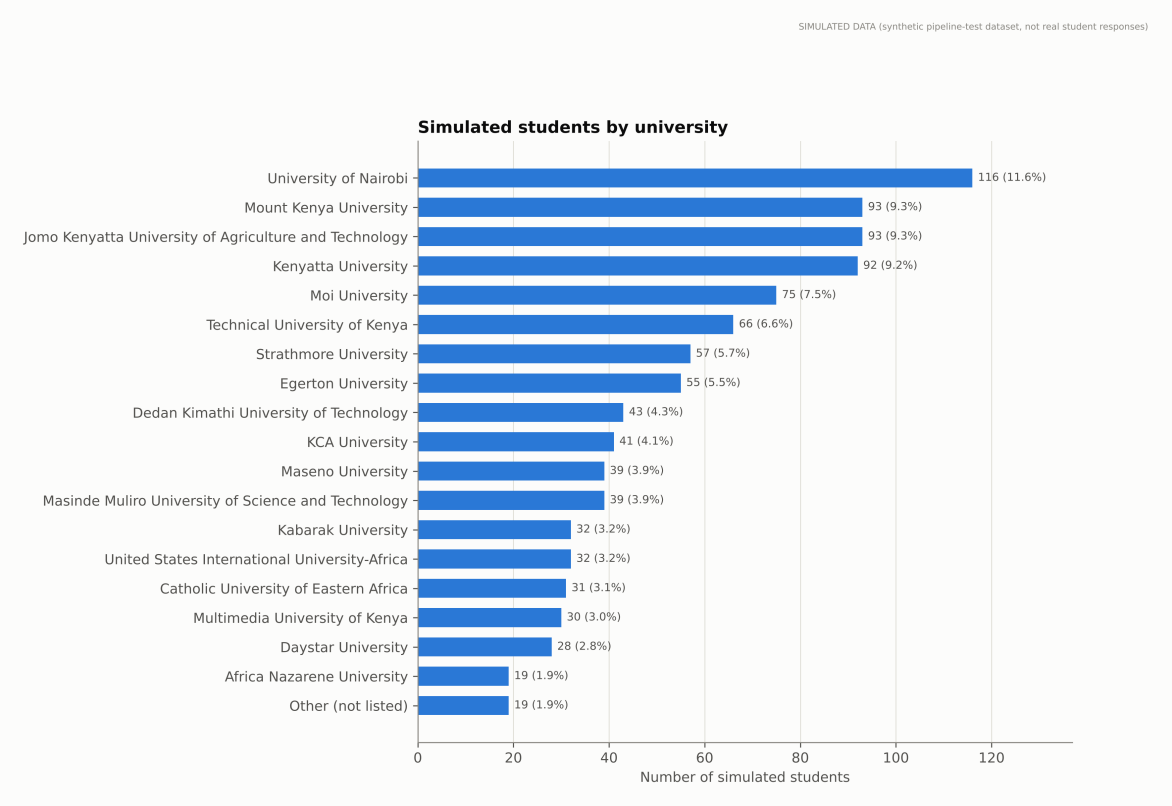


Figure 1. Students by university. SIMULATED DATA.

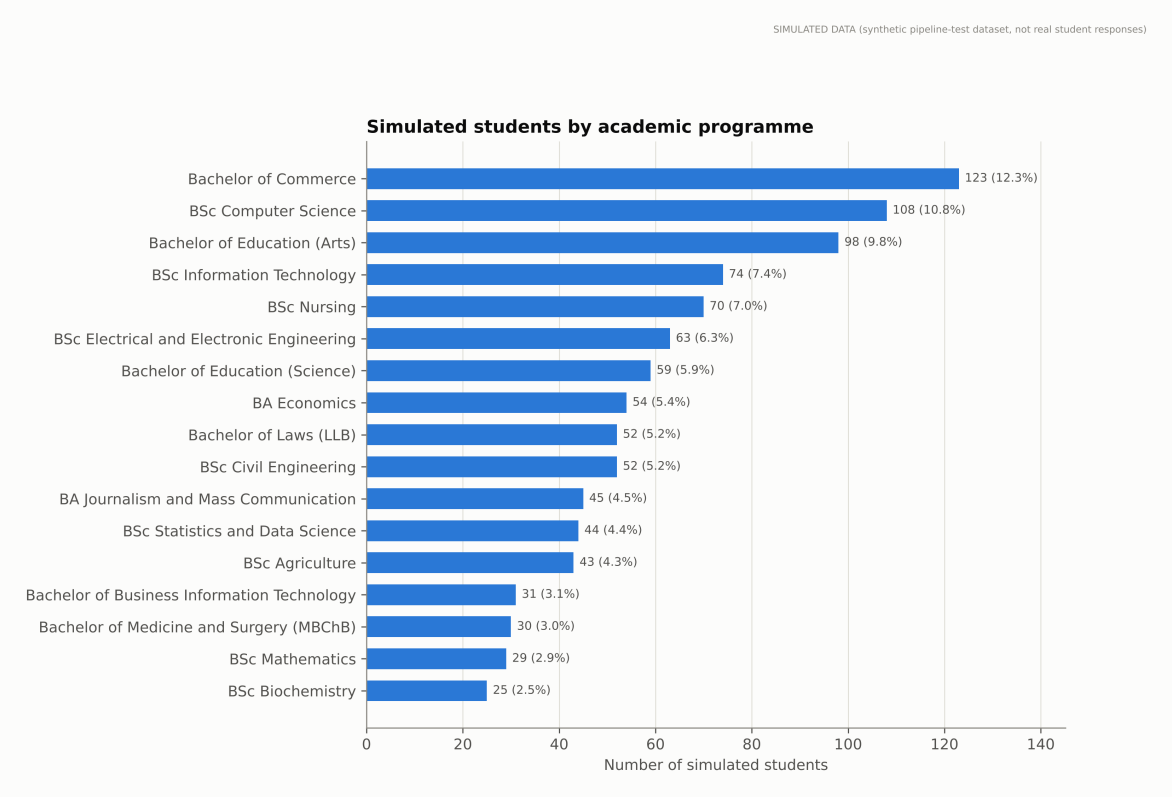


Figure 2. Students by programme. SIMULATED DATA.

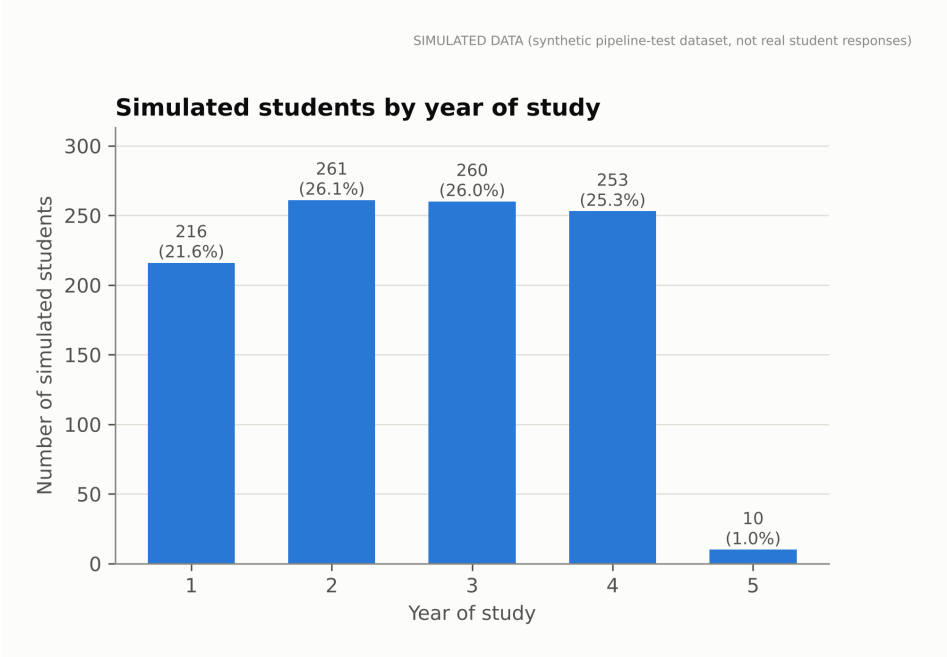


Figure 3. Students by year of study. SIMULATED DATA.

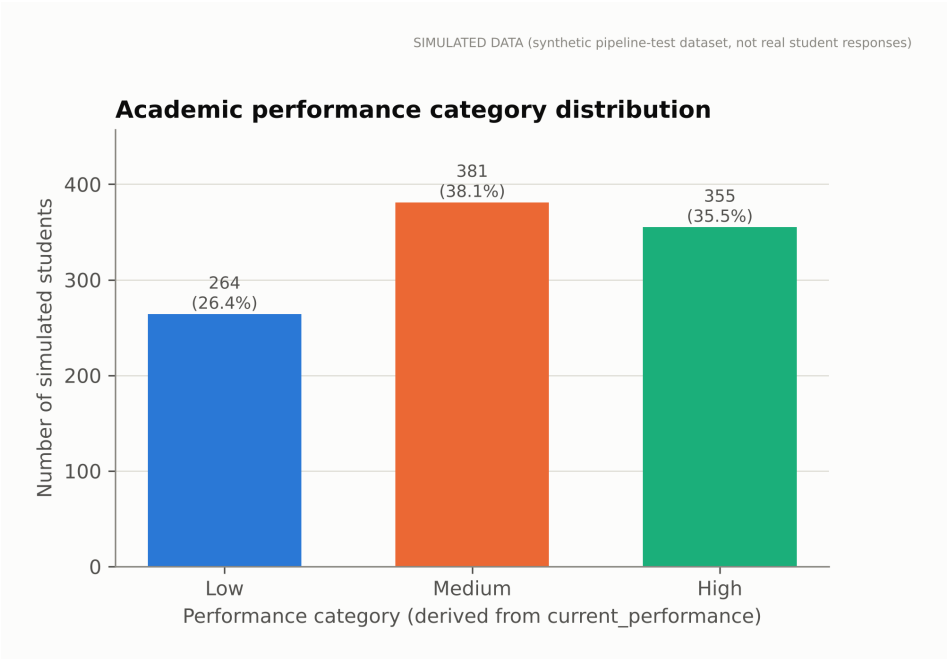


Figure 4. Performance category distribution. SIMULATED DATA.

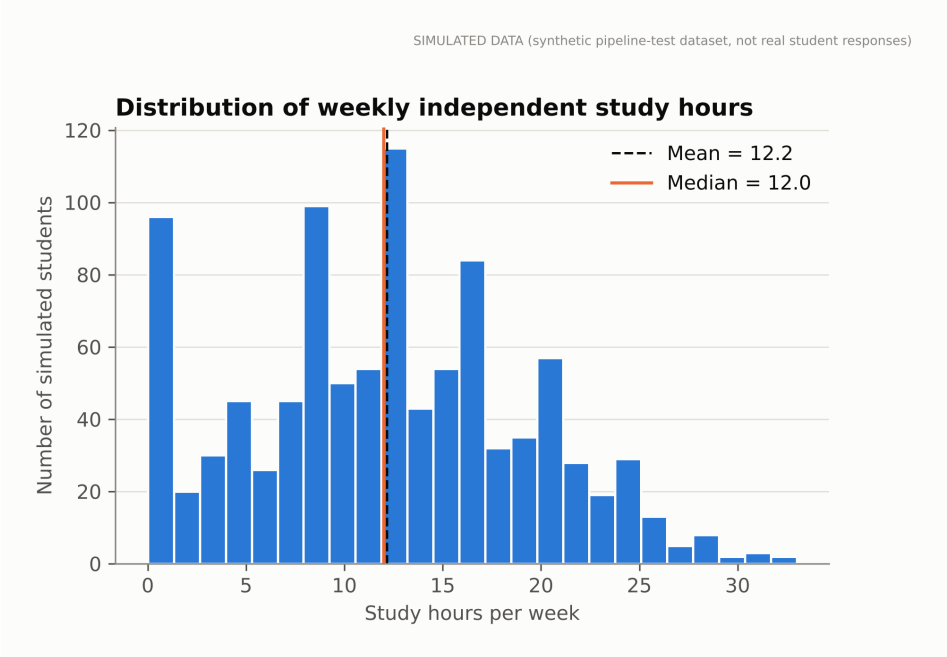


Figure 5. Study hours. SIMULATED DATA.

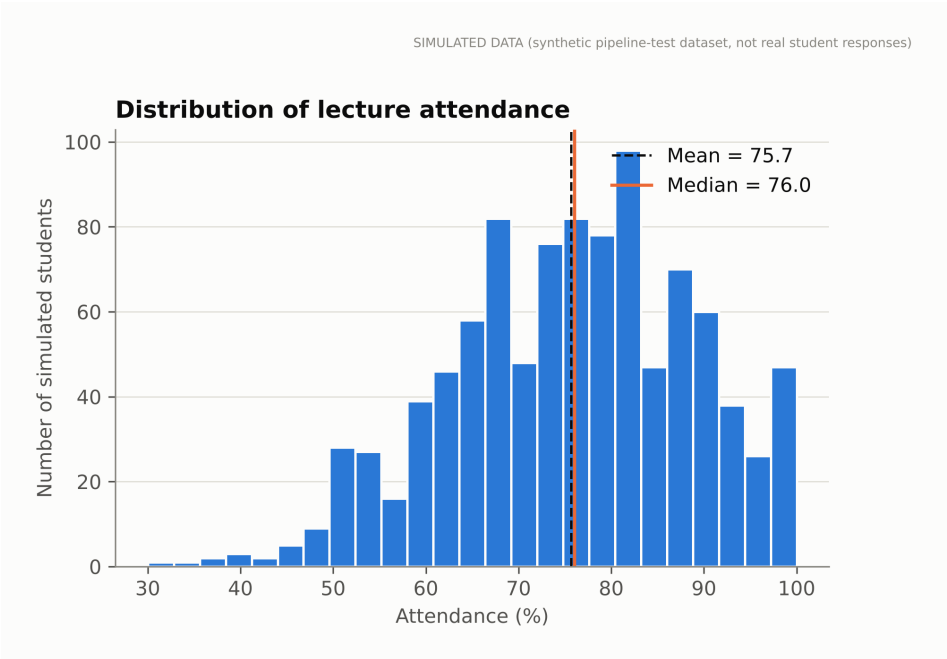


Figure 6. Attendance. SIMULATED DATA.

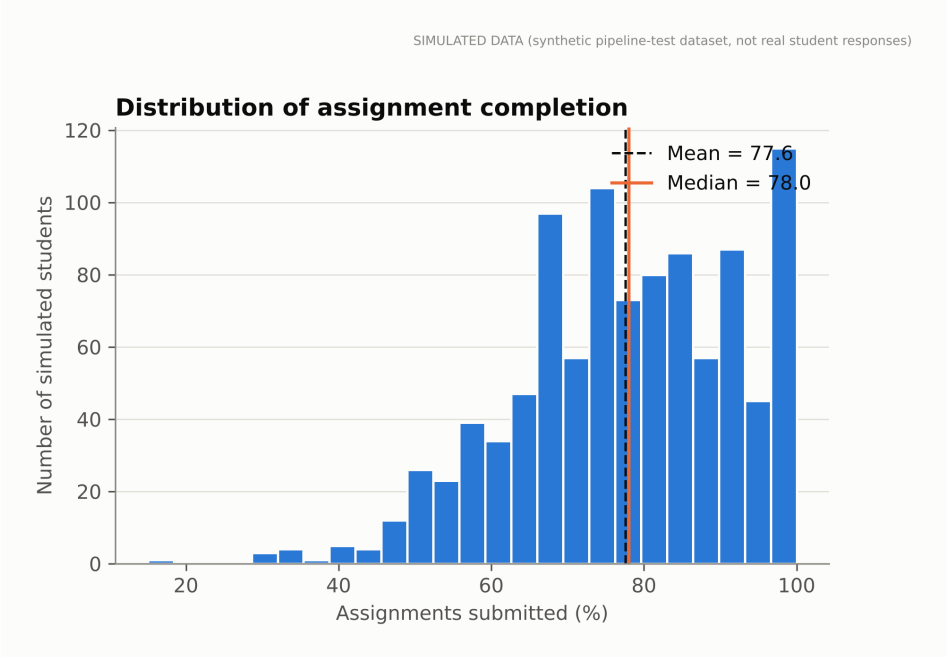


Figure 7. Assignment completion. SIMULATED DATA.

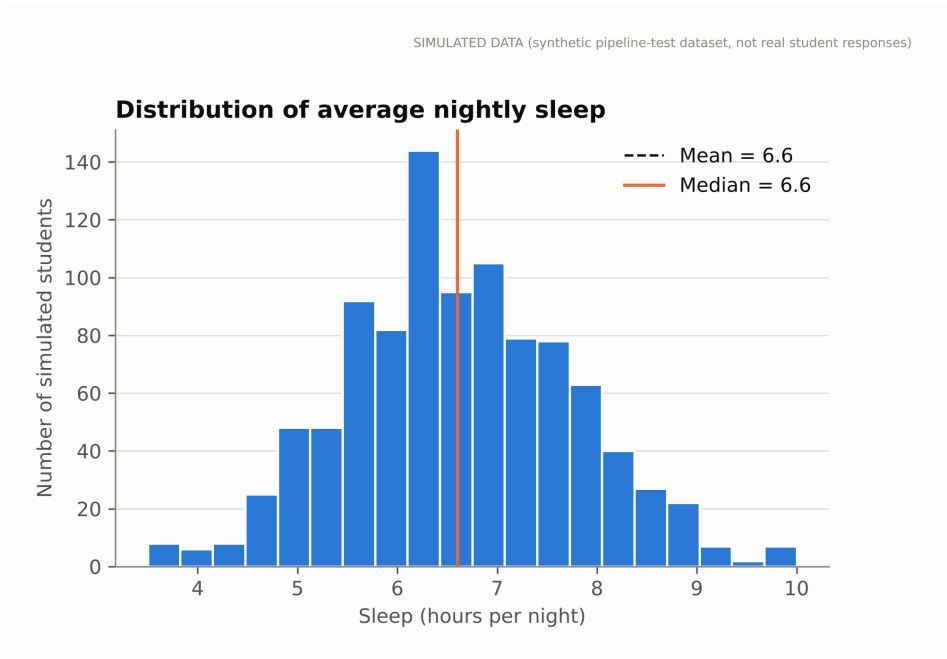


Figure 8. Sleep duration. SIMULATED DATA.

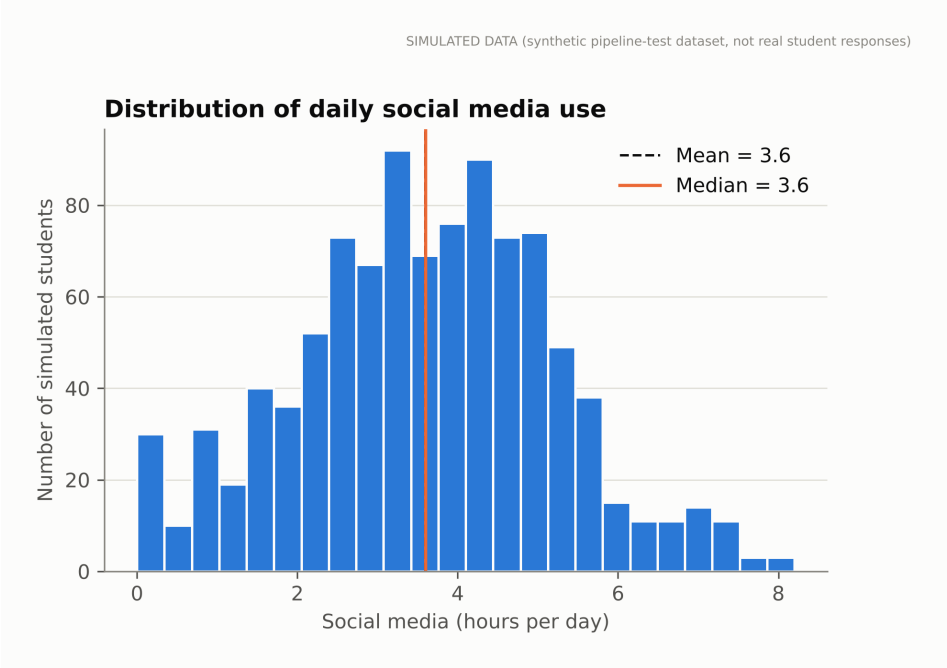


Figure 9. Social media use. SIMULATED DATA.

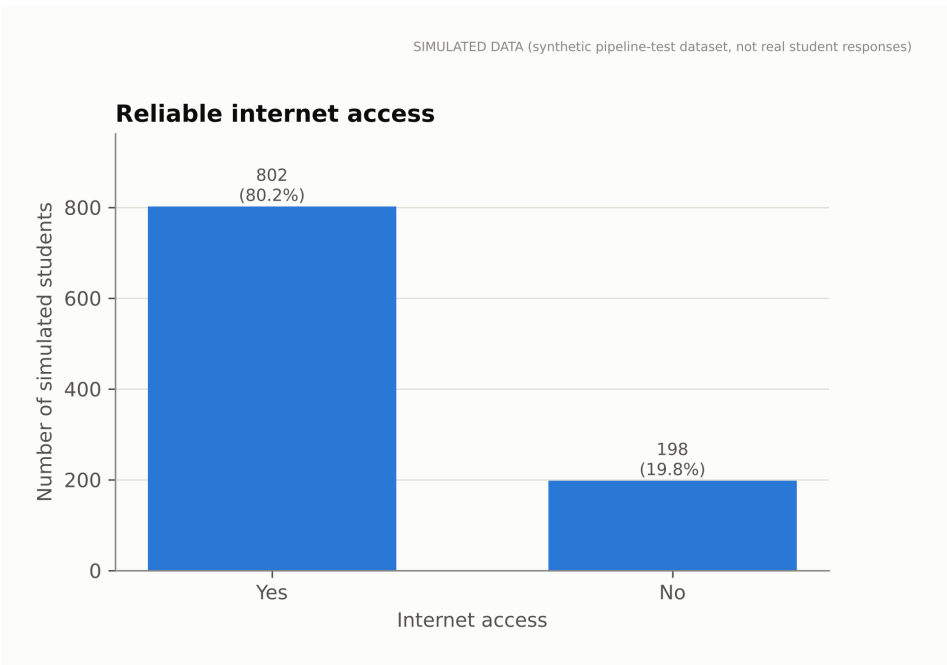


Figure 10. Internet access. SIMULATED DATA.

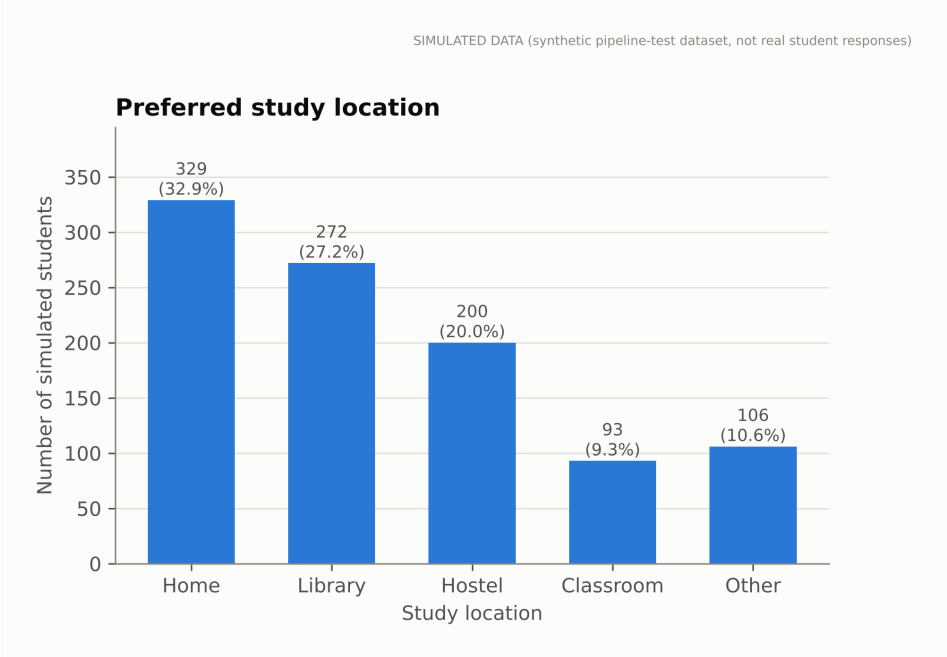


Figure 11. Study location. SIMULATED DATA.

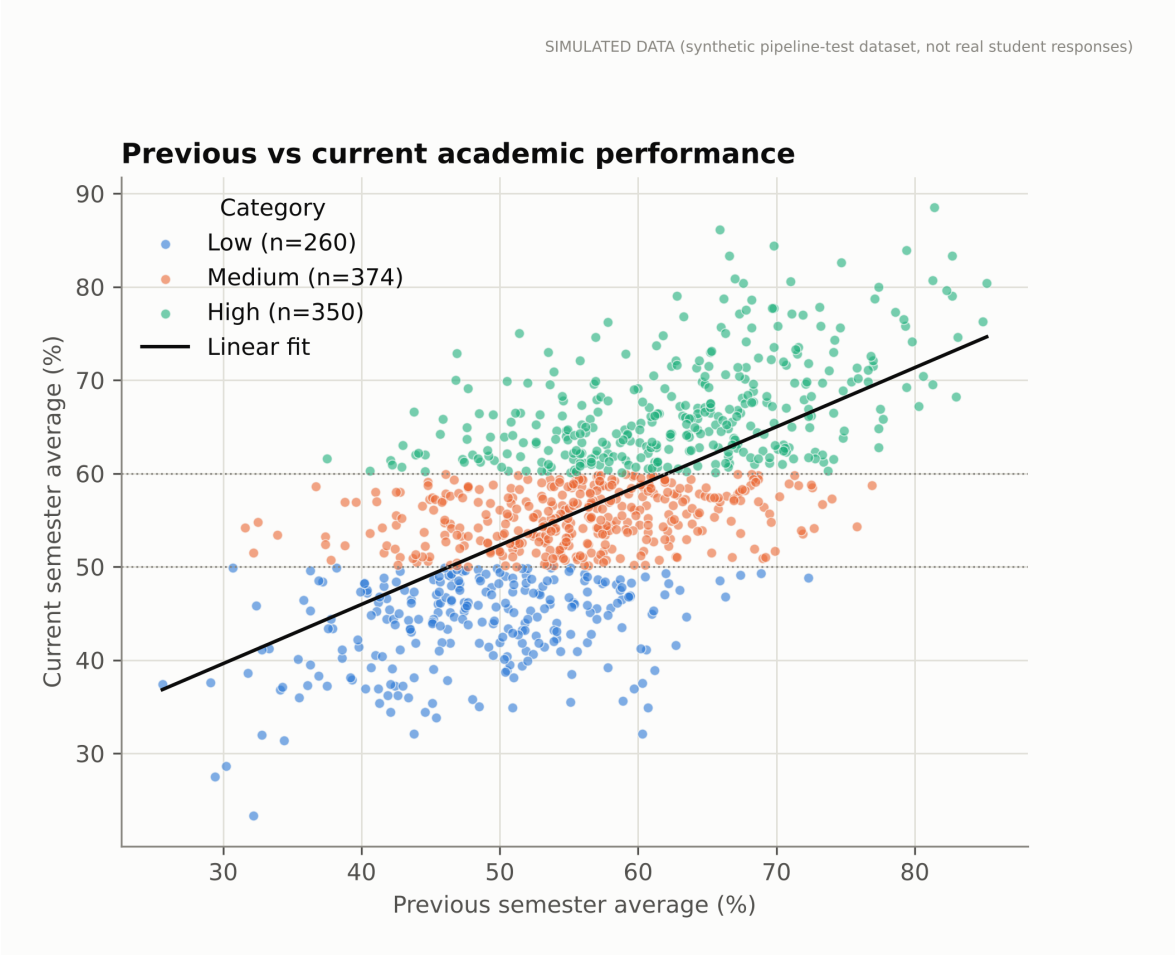


Figure 12. Previous vs current performance. SIMULATED DATA.

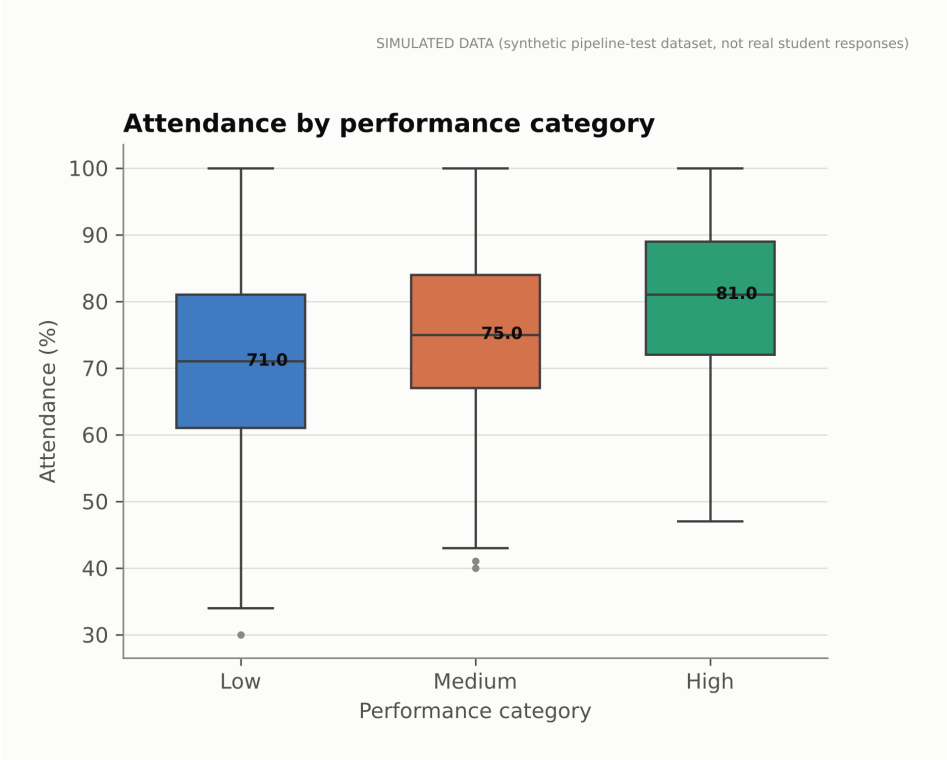


Figure 13. Attendance by category. SIMULATED DATA.

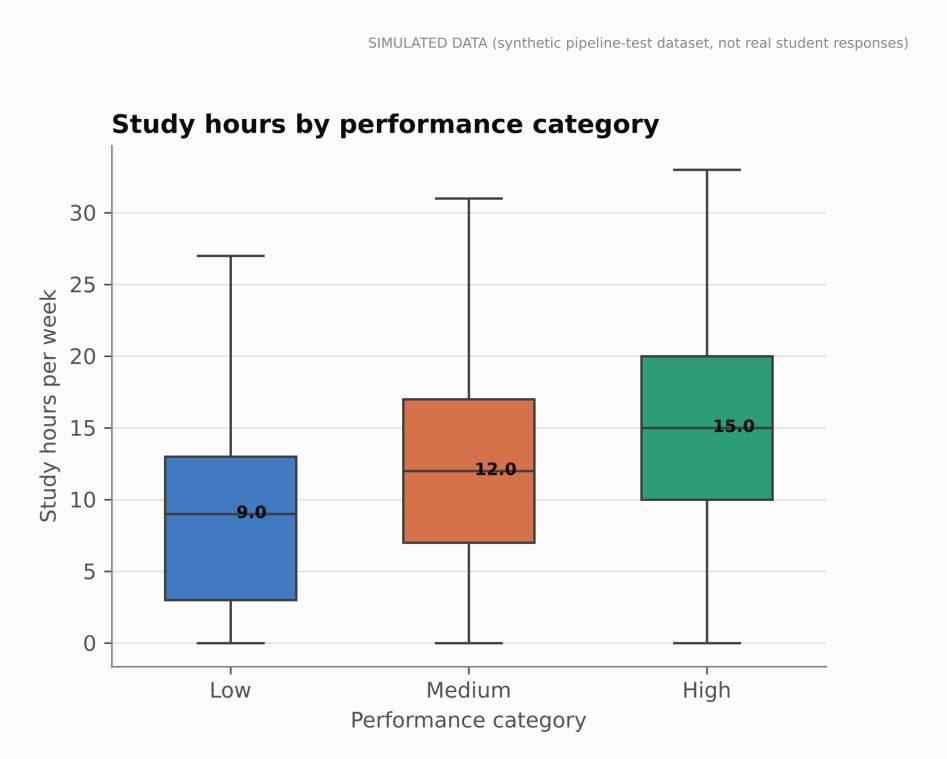


Figure 14. Study hours by category. SIMULATED DATA.

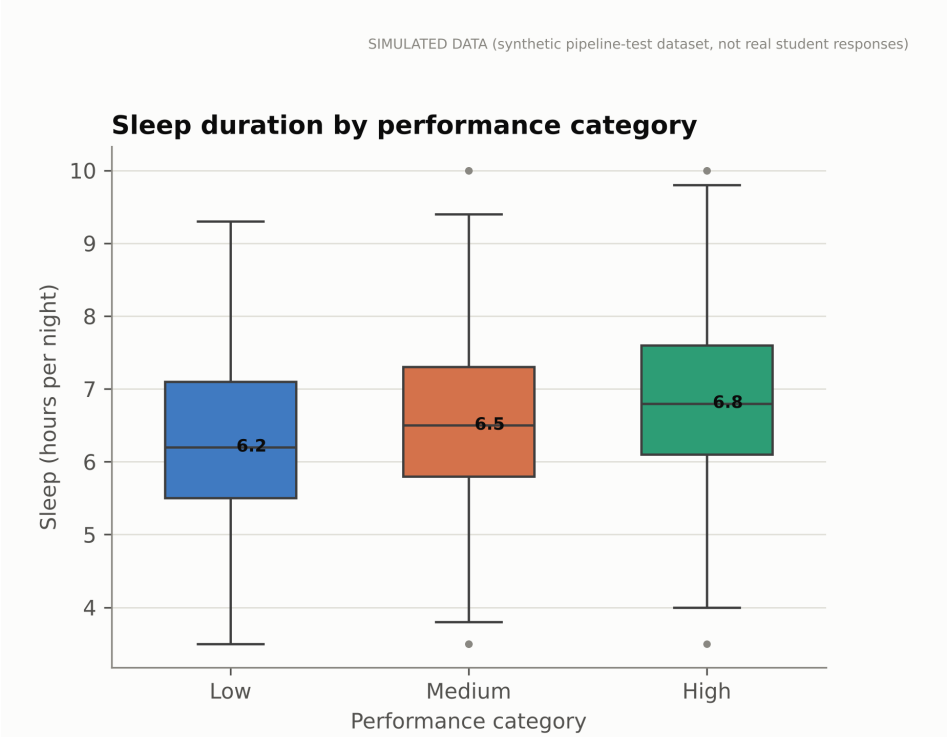


Figure 15. Sleep by category. SIMULATED DATA.

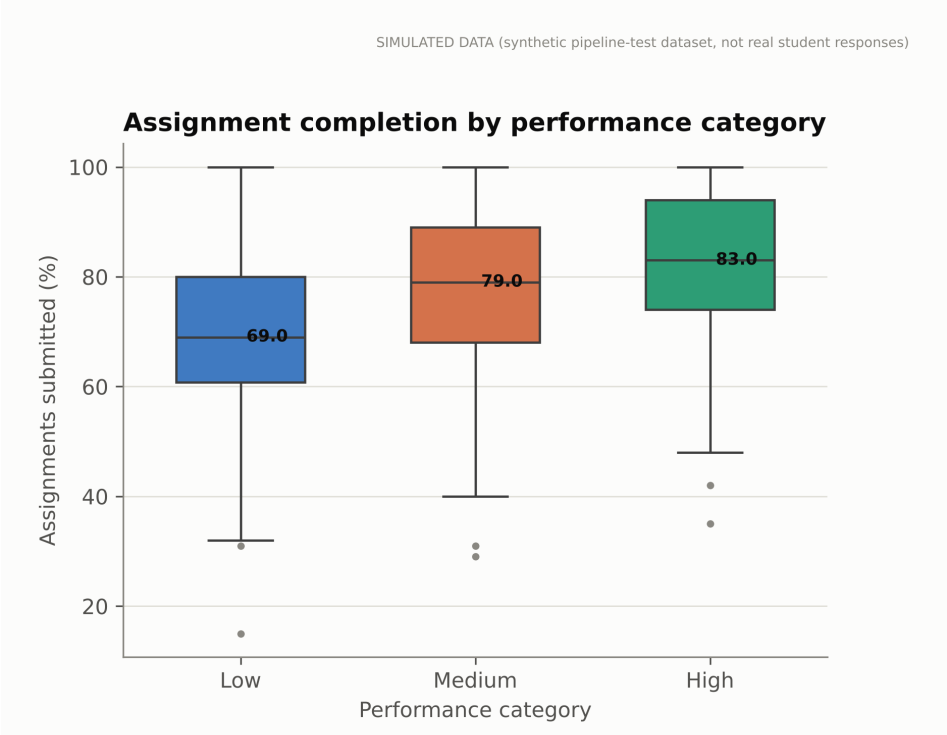


Figure 16. Assignment completion by category. SIMULATED DATA.

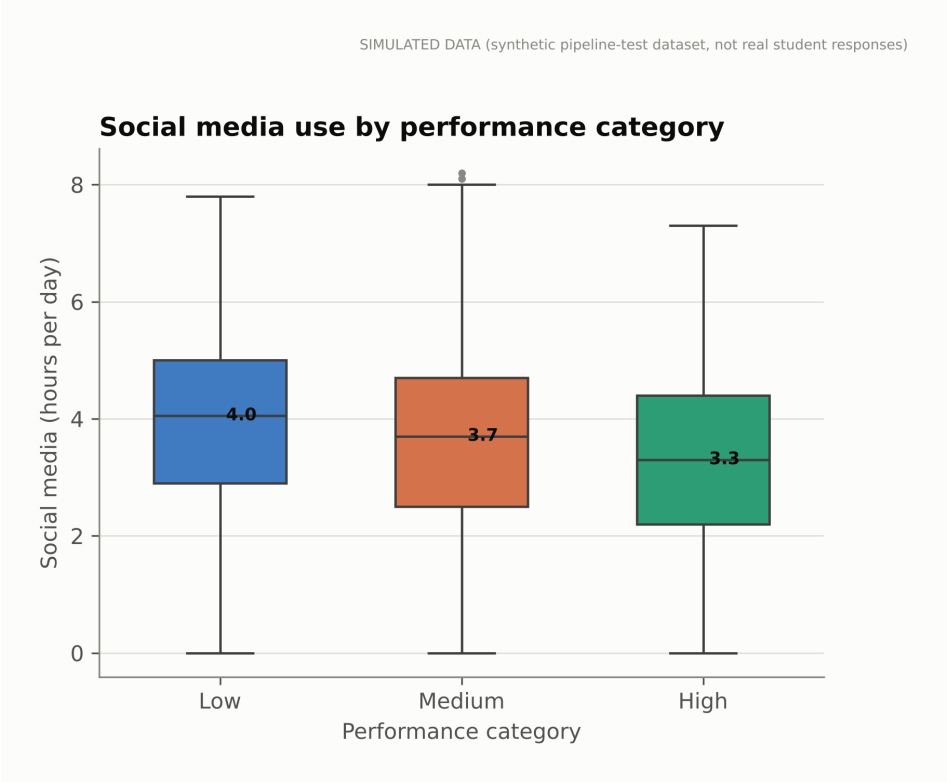


Figure 17. Social media by category. SIMULATED DATA.

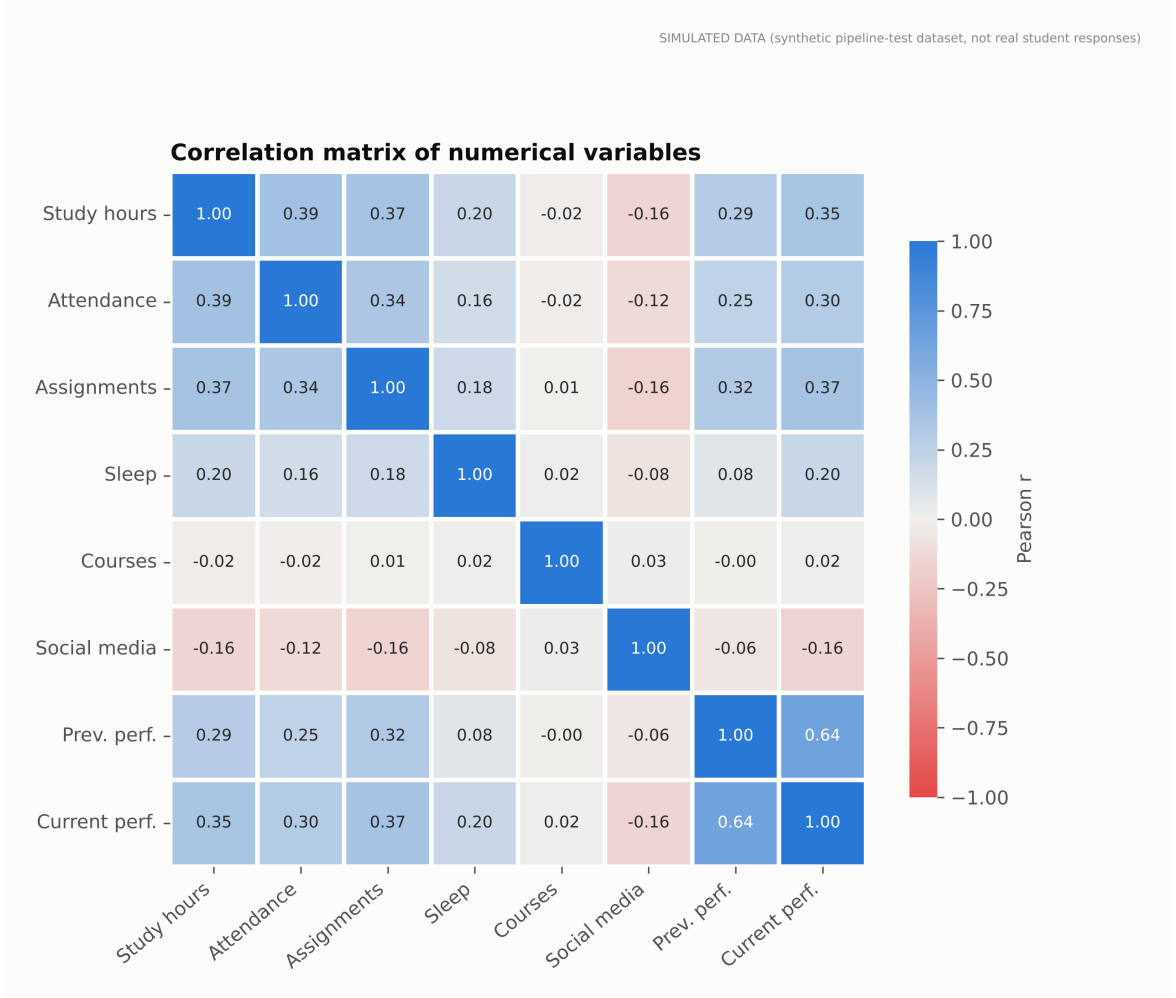


Figure 18. Correlation heatmap. SIMULATED DATA.

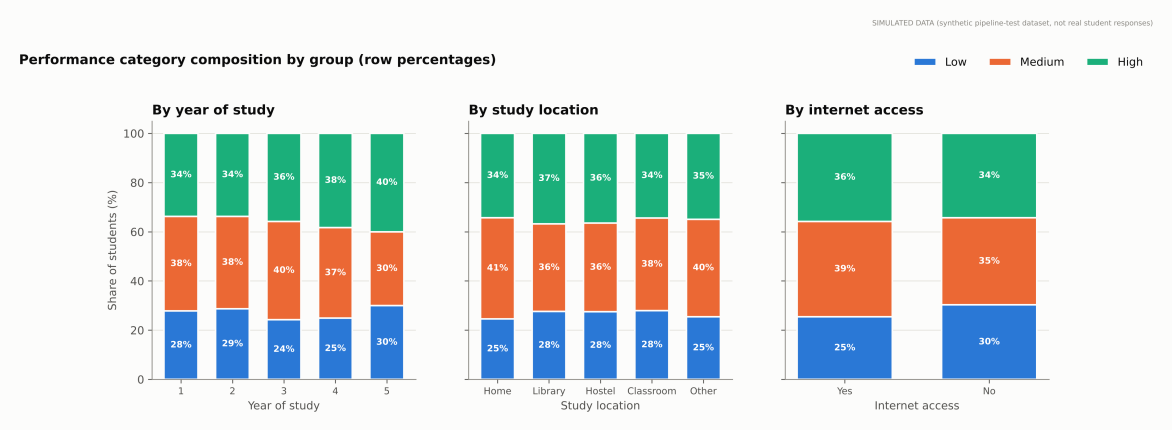


Figure 19. Category composition by year, location and internet access. SIMULATED DATA.

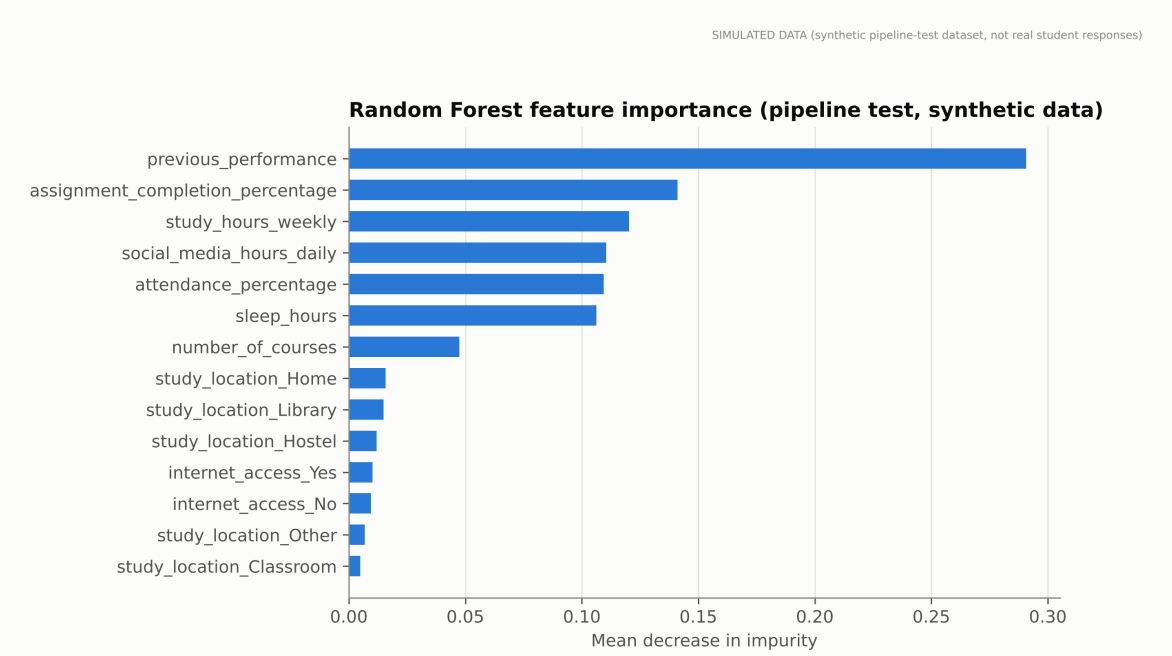


Figure 20. Random Forest feature importance (pipeline test). SIMULATED DATA.

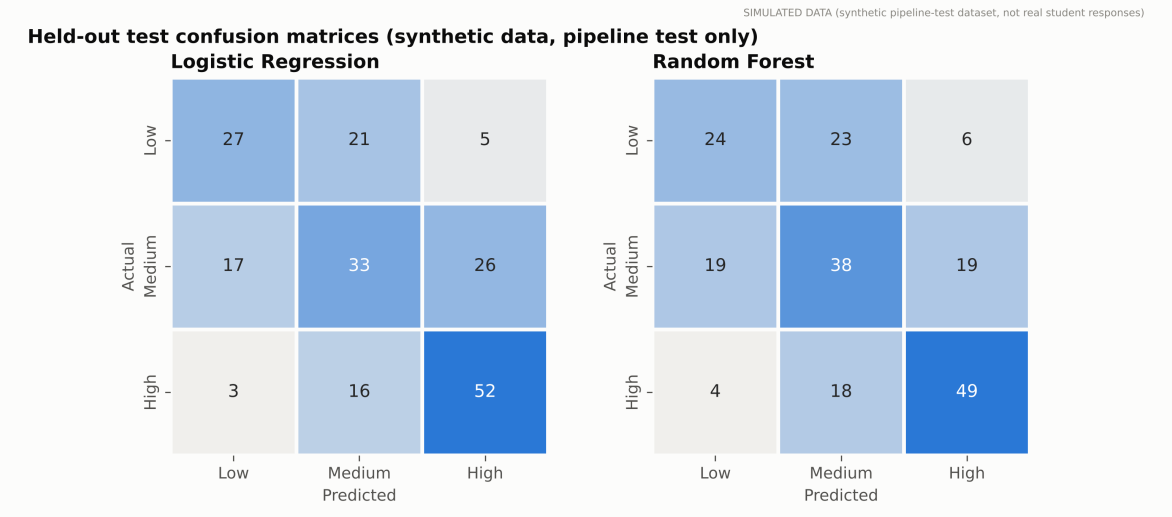


Figure 21. Test confusion matrices (pipeline test). SIMULATED DATA.